Nomodynamics: The Mathematics of Self-Amending Law

A founding note

The Treeline Program

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Abstract

Game theory formalized games with fixed rules; Conway's deletion of the *players* gave cellular automata and the Game of Life. The orthogonal deletion—removing the *fixity of the rules*—was never taken, although rules that amend rules are humanity's oldest formal practice. We introduce a minimal, parameter-free mathematical object native to that vacancy: a system whose only substance is laws acting on laws. Its founding ("own-kind") semantics turns out to be exactly solvable, with short theorems that carry immediate institutional readings: fully occupied codes are frozen (*gridlock*); stable codes are always fully blocked (*dead letters*); and on the integer line *the eldest law cannot be repealed*, which makes free-moving law-packets impossible—while on circular codes, which have no eldest law, moving packets provably exist. The solvable sector exhibits a small charismatic fauna (colonizers, sunset clauses, welds, refraction, rotating packets, and an aperiodic avalanche machine locked to powers of two). Letting laws amend *other* laws then settles what motion costs: a tropical monovariant shows that a constitution in which every law amends a single kind—anyone's kind—is motionless in every dimension, window and resolution, while two targets suffice, and the smallest traveller is one placed law. The same generalization splits the stability theorem by resolution convention and produces *balanced constitutions*: codes perpetually active and perfectly unchanging.

1 The vacancy

Three observations, from a systematic survey of how new mathematical ontologies have historically been found, converge on one empty cell. (*i*) The deletion grid: von Neumann–Morgenstern formalized 2-player fixed-rule games; puzzles are the 1-player case; Conway's "0-player games" are cellular automata. Along the orthogonal axis—rule fixity—no elementary mathematics exists. (*ii*) The reduction shadow: computability theory's founding reduction, "without loss of generality the program is fixed," is valid for *what can be computed* (universality lets a fixed machine simulate a self-modifying one) but silently discards the *behavioral* questions: what do small self-amending systems actually do? Nobody looked, because the reduction said one need not. (*iii*) The practiced verb: constitutions and statutes—self-amending by essence—are the oldest running formal systems, yet "amend" was never objectified the way Church and Turing objectified "compute." The fragments that exist (von Neumann's rule-carrying constructor cells; Suber's game *Nomic*, 1982; self-modifying Petri nets, 1978; algorithmic chemistry) never produced a minimal canonical object, a census, or a theorem.

2 The object

Definition 1 (nomic chain)**.** A *law* is a triple $(a, b, c) \in \{-1, 0, 1\}^3$ placed at a position $i \in \mathbb{Z}$. A *code* is a finite set S of placed laws. A placed law $(i, (a, b, c)) \in S$ is *active* if some law stands at $i + a$ and no law stands at $i + b$. Time is synchronous: every active law flips the presence of *its own kind* at $i + c$, flips resolving by parity. Nomic *rings* are the same system on $\mathbb{Z}/m$; the d-dimensional and window-w versions replace $\{-1, 0, 1\}$ by offset sets in $\mathbb{Z}^d$.

There is no board and no external rule table: the law is the only substance. The system is parameter-free. All claims below were pre-registered as predictions or risks before the first simulation, then settled by censuses with machine-checked certificates ($\sim$15.3 million certified runs for the no-go results; complete enumerations where stated).

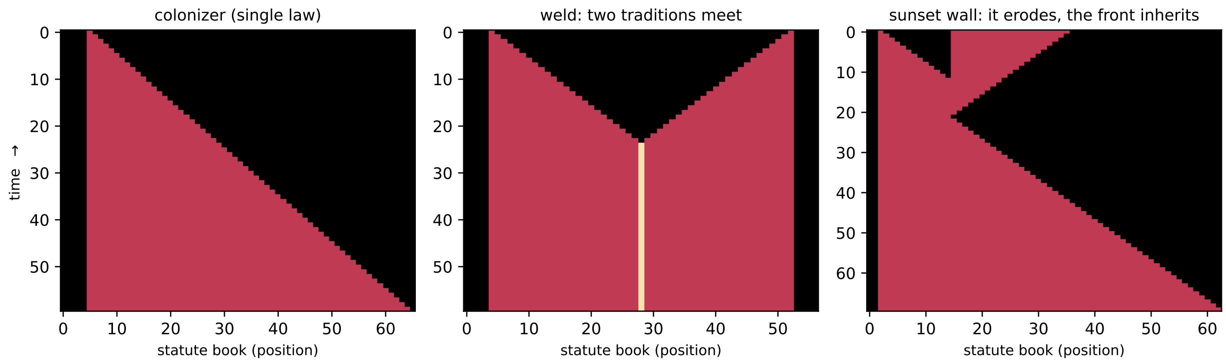

Figure 1: Window-1 nomic chains (time downward). Left: the *colonizer* $(0,1,1)$—“while I stand and my right is vacant, enact my kind to the right”—only its frontmost copy is ever active. Middle: two traditions meet and *weld* into frozen code with a double-law seam. Right: a colonizer meets a self-eroding *sunset wall*. The wall repeals itself from its far end while the front stands blocked; the front then fills the vacated ground, so a wall of length L is cleared at time exactly $\max(2L,\, x_0+L)$ — the “refraction index 2” is those two speed-1 processes in sequence. The colonizer never converts the wall: it waits for the old code to sunset and takes the vacancy.

3 First theorems: the solvable sector

Theorem 2 (Gridlock)**.** *In a fully occupied region every law's vacancy guard fails; a solid code is frozen. All dynamics is surface dynamics.*

Lemma 3 (Single Author)**.** *The only law that can ever flip kind $k=(a,b,c)$ at position j is the kind-k law at $j-c$. Hence parity and idempotent (“OR”) resolution coincide identically, in every dimension, and per-kind dynamics is occupancy-modulated linear algebra over $\mathbb{F}_2$.*

Theorem 4 (Dead Letter)**.** *A code is a fixed point iff every law in it is blocked. In particular* balanced constitutions*—codes in perpetual self-cancelling activity—do not exist: stability is always gridlock.*

Theorem 5 (Anchor)**.** *On $\mathbb{Z}$ (any window, any dimension, any guard predicate, either resolution semantics), own-kind toggles of kind t land only at t's own offset c_t; consequently the extremal law on the trailing side of a finite code is never targeted:* the eldest law cannot be repealed*. No finite code satisfies $\Phi^p(S)=\sigma^d(S)$ with $d\neq 0$: free gliders are impossible.*

Corollary 6 (Ring rotors)**.** *On $\mathbb{Z}/m$ there is no eldest law, and the obstruction genuinely vanishes: the minimal moving law-packet of nomodynamics is three laws of the single kind $(0,1,-1)$ at cells $\{1,2,5\}$ of $\mathbb{Z}/6$, hopping $m/2=3$ cells per step (Fig. 2); such rotors exist on every even $m\geq 6$ and on no odd ring ($\leq$3-law sweeps complete for $m\leq 7$).* Entrenchment is a theorem of linear order: linear statute books are anchored by their oldest provision; circular codes can revolve.

Remark 7. The lemma explains the observed clockwork: every exact cycle found across ∼140,000 seeds has period a power of two, and single-kind ring codes (the “Sunset Parliament”) attain exact maximal periods 15, 63, 341 at resonant circumferences $m\equiv 2 \pmod 4$—orders of $\mathbb{F}_2$-polynomials governing constitutional cycles.

4 Fauna of the solvable sector

The one-dimensional chain is a calculus of fronts (Fig. 1): *colonizers* (2 of the 27 kinds) fill the line at speed 1; the *sunset clause* $(0,-1,1)$ forever enacts a subordinate its own presence then blocks (period 2); *sunset codes* dissolve from their newest end; colliding fronts *weld*; porous codes support one-way *conversion waves* (speed 2/3); walls facing a front sort into opaque (19/27), pulsing (3/27), and effectively transparent (5/27) classes. In two dimensions a *ray-confinement theorem* pins each kind to one axis ray (quadrants never fill), and the fauna adds 239 certified half-turn *pinwheel rotors*, Pascal-column growers with $|S_t|=2^{\mathrm{popcount}(t)}$, and

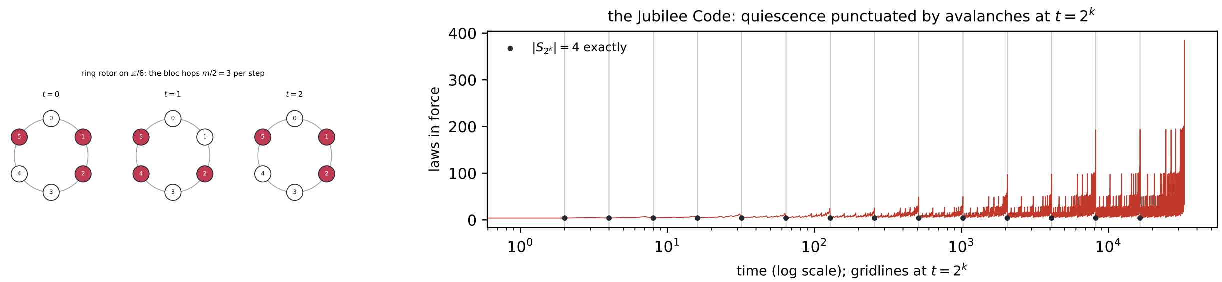

Figure 2: Left (2): the minimal ring rotor. Right: the *Jubilee Code*, a ∼26-law two-dimensional machine, aperiodic through 300,000 fully-hashed steps and quiescent except at $t = 2^k - 1$, when the counter fills and the whole code ignites; at $t = 2^k$ itself exactly four laws stand, every time (16/16, complete through $t < 2^{17}$), and the crest height doubles every second power of two. A binary counter native to law-space, and an attractor: 791 of 60,000 random seeds converge to its family.

the Jubilee Code (Fig. 2, right). Finite rings admit an exact fixed-point count (transfer matrix, brute-force verified), a predecessor algebra with Garden-of-Eden fraction peaking ≈ 33% at mid-density, and a unique minimal taut architecture (the two-law *mutual-veto* pair).

5 Cross-amendment: what motion costs

Own-kind amendment is the tame simplification; real statutes amend other statutes. Generalize minimally: a *constitution* is a finite kind set K, each $k \in K$ carrying a rule (a_k, b_k, c_k) and a *target set* $T_k \subseteq K$; an active law of kind k at i toggles every kind of T_k at $i + c_k$. Own-kind is $T_k = \{k\}$. Nothing external is added: out-degree 1 makes the amendment digraph a functional graph, so a kind is a pointed finite graph with a triple at each node—*"I am (a, b, c), and I amend the law that is (a', b', c') and amends…"*—and own-kind laws are exactly the self-loops. The law is still the only substance.

The expectation was that cross-kind targeting is the escape from the Anchor Theorem. It is not.

Theorem 8 (Out-Degree Law)**.** *If, restricted to the kinds a pattern uses, every law amends* at most one *kind, no free glider exists—any offsets, any window, any dimension, parity or OR. This contains the Anchor Theorem (self-loops), reciprocal amendment, and every permutation constitution of every cycle length.*

The proof is a tropical monovariant: with weights w satisfying $w_k - c_k \le w_{t(k)}$, the quantity $\Psi = \min_t(\alpha_t + w_t)$ over the kinds' trailing edges can rise only through a tight cycle of weight 0. The same argument bounds speed.

Theorem 9 (Tropical Speed Law)**.** *For a glider of period p and displacement d, $p \min(\lambda_{\min}, 0) \le d \le p \max(\lambda_{\max}, 0)$, where $\lambda_{\min}, \lambda_{\max}$ are the extreme cycle means of the amendment digraph (edge $k \to m$ of weight c_k for each $m \in T_k$). If every cycle has weight zero, or the digraph is acyclic, nothing moves.*

Theorem 10 (Supersession no-go)**.** *State-dependent* supersession *targeting—enact your own kind on empty ground, clear the whole cell if occupied—admits no free glider in any dimension: its creation channel is still own-kind, so every present kind must push forward, and then nothing can clear the rearmost cell.*

The threshold is exact: out-degree 2 suffices and the specimens are tiny (Fig. 3). The cleanest evidence is a controlled experiment—three semantics sharing their guards, their offsets and their destruction rule exactly and differing only in the out-degree of the *creation* channel—giving 0 gliders in 1,119,744 complete classifications at out-degree 1 and 3,424 certified gliders in 279,936 at out-degree 2. So *cross-amendment was never the obstruction: narrowness was.* A law that amends one provision, even somebody else's, cannot move; a law that amends two can. Entrenchment, in chapter one a consequence of linear order, is equally a consequence of legislative narrowness.

The same degree condition governs *growth*. Own-kind two-dimensional growth is pinned to axis rays ($\alpha \le 1$, a theorem of chapter one); the generalisation is exact and cheap: out-degree 1 forces $|S_t| \le |S_0|(t + 1)$—hence

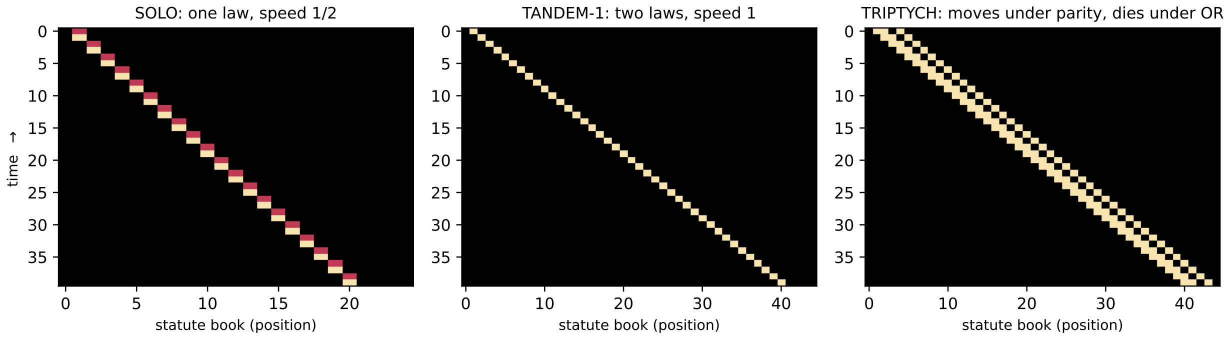

Figure 3: The first free gliders of nomodynamics (time downward). *SOLO*: one placed law travelling forever at speed $\frac{1}{2}$ under a fully-cross constitution. *TANDEM-1*: two laws in a *single* cell, period 1, speed 1—the maximum; it lifts to $\mathbb{Z}^2$ at any velocity, knight moves included, and revolves on every ring $m \geq 3$, where own-kind rotors need even $m \geq 6$. *TRIPTYCH*: a glider under parity that under OR does not travel at all but detonates into a lattice growing by three laws per step.

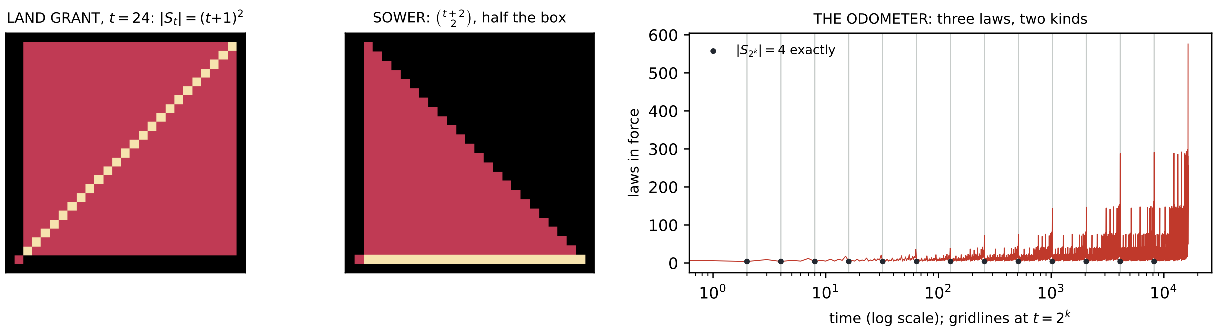

Figure 4: Two dimensions. Left: *Land Grant*—one law with two targets, and $|S_t| = (t+1)^2$ exactly, a solid growing square: the plane fills, which own-kind ray confinement forbids. Middle: *Sower*, two laws, exactly half the box. Right: *the Odometer*, three laws of two kinds, a second binary counter—quiescent, avalanching at $2^k - 1$, and standing at exactly four laws at every $t = 2^k$ (checked to $t = 2^{20}$), with reach growing like $(\log t)^2$: bounded population, unbounded memory, never repeating.

$\alpha \leq 1$ in every dimension, under parity, OR and supersession alike—while out-degree 2 attains $\alpha = 2$ from a single placed law (Fig. 4). Ray confinement was never a fact about two dimensions; it was a fact about narrow laws. The degrees of the amendment digraph turn out to govern the whole zoo: out-degree ≤ 1 forbids motion and caps growth; out-degree ≥ 2 buys gliders and area; a kind of *in-degree zero*—a provision nobody can amend—is a pump, and yields guns and rakes, on $\mathbb{Z}$ as readily as on $\mathbb{Z}^2$. *An entrenched clause is a gun.*

Multi-authorship also revives the resolution axis that the Single-Author Lemma had made vacuous, and it kills the Dead Letter Theorem in exactly one convention.

Theorem 11 (Balance)**.** *Under parity, two active laws whose enactments cancel leave a code fixed forever while remaining alive—a* balanced constitution*; the minimal witness is two placed laws. Under OR no such code exists at any size: a fixed point still requires every law blocked, and Dead Letter survives verbatim.*

A constitution can be perpetually active and perfectly unchanging— but only where simultaneous contradictory amendments annul one another instead of both taking effect. Balance is governed by the *other* degree: a provision with two authors. It is not a curiosity of small codes—balanced codes of unbounded size exist with *every* law active at every step, their exact count satisfies $a(s) = 4a(s-1) - 2a(s-2)$, and 70% of the balanced verdicts in a complete 143-million-run census are reached rather than seeded.

Finally, the Anchor Theorem itself has a deeper reading than "own-kind amendment forbids motion". A sweep of the semantics the referent suggests—entrenchment clauses, quorum guards, later-law-wins, sunset-by-default—finds that all but one leave the confinement theorems untouched (they are guard-free), while *sunset-by-default*, where a law lapses unless re-enacted, admits free gliders on $\mathbb{Z}$ with plain own-kind targeting: 11–15% of a complete 139,968-code box, at speeds 1, $\frac{2}{3}$, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{6}$. So there are exactly two ways known to make a statute book move:

legislate broadly (out-degree ≥ 2), or let provisions expire on their own.

What forbids motion on the line is **permanence**, not own-kind targeting—the Anchor Theorem's real content.

6 The frontier, located by theorem

The theorems above are cheap to state and were found within two days of the definition, which is the usual sign that a territory has been entered rather than exhausted. Open problems, in the order we would attack them. (1) *Speed quantisation, and a warning about boxes.* The apparent law "the number of kinds caps the speed" is an artefact, and the way it failed is worth more than the law would have been. Bounded searches decide a *box*; every census in this program used boxes of at most ∼26 interior cells, and the two-kind universe Mirror ($W = 1$, rules $(0, 1, -1)$ and $(0, -1, 1)$, each amending both kinds) carries displacement-2 gliders of minimal period $4, 5, 6, 7, 12$ with spans 53, 20, 616, 438, 39—all invisible to those boxes, and all certified. A second confound compounds it: sweeps over *coprime* (p, d) are structurally blind to $\Phi^4 = \sigma^2$ when $\Phi^2 = \sigma^1$ fails. So every "decided impossible" in a fixed box, here and elsewhere, means *no narrow glider.* What does survive is sharper: in the single-field sector (all kinds sharing the same target set, so the system is a one-bit automaton with n channels) the measured cap is $|d| \leq 2$ under parity and $|d| \leq 1$ under OR, *at any width and any number of channels*; the resource that buys $|d| \geq 3$ is another $\mathbb{F}_2$ field, not another kind. Prove that cap, and explain why it fails to scale with W: at $W = 2$ and three kinds displacements reach at least $5 > 2W$, while a dilation $(W, p, d) \mapsto (rW, p, rd)$ rules out any width-independent bound a priori. (2) Is any small cross-amendment universe computation-universal? The gate-level inventory—what reflects, what annihilates, what survives collision—comes first; the claim comes much later, if at all. (3) Classify ring-rotor families (which m, which hop groups) and the period spectrum: own-kind periods are locked to powers of two by linearity, and multi-authorship should break that lock. (4) Prove the Jubilee clock law—$|S_t| = 4$ at *every* power of two, crest at $2^k - 1$ doubling every second power—and explain its basin: 791 of 60,000 random seeds fall into this one attractor. (5) The reception question the founding practice suggests: which theorems survive into semantics faithful to actual amendment procedure—entrenchment clauses, quorum guards, later-law-wins, sunset by default?

Methods. Definition and predictions were frozen before the first run; censuses carry machine-checked certificates (recurrence, translation-recurrence, rotation-recurrence); complete enumerations: 1- and 2-law seeds (W=1), ≤4-law/5-cell patterns (8.1M), ring sweeps $m \leq 12$; the Anchor and Dead Letter theorems have short combinatorial proofs checked against 15.3M runs with zero violations. Full records: the Treeline program archive, `phase6/`.